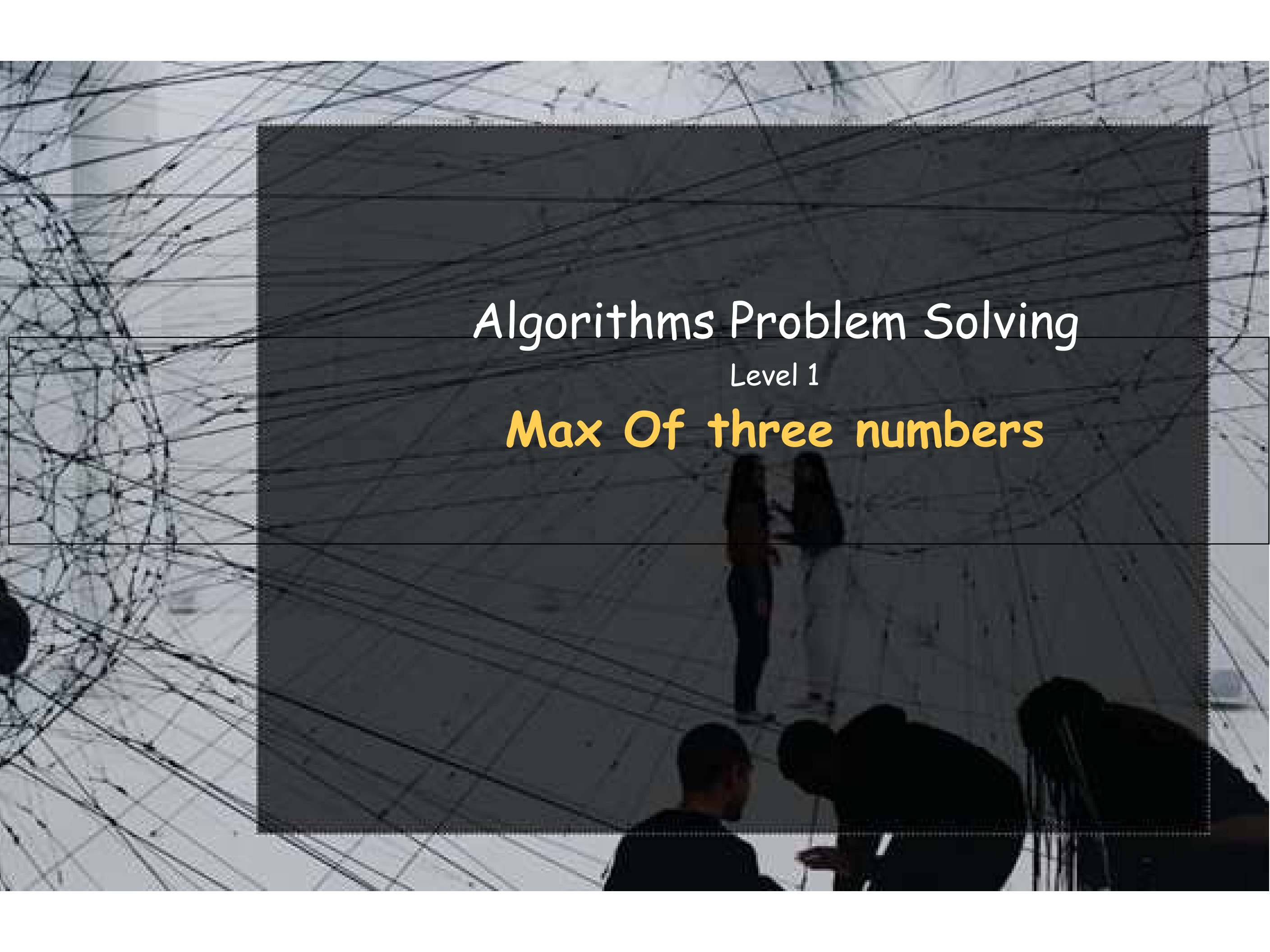


Algorithms Problem Solving Level 1

Max Of three numbers



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Definition

- ◆ This program reads three numbers from the user and finds the maximum value among them.
- ◆ The program uses functions, reference parameters, if-else conditions, logical AND, and return values.

Explanation

- ◆ The `readNumbers()` function reads three numbers and stores them in `num1`, `num2`, and `num3`.
- ◆ The parameters `int& num1`, `int& num2`, and `int& num3` are reference parameters, so the function can directly modify the variables from `main()`.
- ◆ The `maxOf3Numbers()` function compares the three numbers:
- ◆ If `num1` is greater than or equal to both `num2` and `num3`, it returns `num1`.
- ◆ Otherwise, if `num2` is greater than or equal to both `num1` and `num3`, it returns `num2`.
- ◆ Otherwise, it returns `num3`.
- ◆ The `&&` operator means that both comparisons in a condition must be true.
- ◆ The `printResult()` function receives the maximum number and prints it.
- ◆ In `main()`, the functions are combined:
- ◆ `readNumbers(num1, num2, num3);`
- ◆ `printResult(maxOf3Numbers(num1, num2, num3));`

Steps in main()

1. Create three integer variables.
2. Read the three numbers using readNumbers().
3. Check if num1 is greater than or equal to num2 and num3.
4. If true, return num1.
5. Otherwise, check if num2 is greater than or equal to num1 and num3.
6. If true, return num2.
7. Otherwise, return num3.
8. Pass the returned value to printResult().
9. Print the maximum number.

💡 Example

◆ If the user enters:

Number 1: 25

Number 2: 40

Number 3: 30

◆ num1 is not the maximum.

◆ num2 is greater than both num1 and num3.

◆ maxOf3Numbers() returns 40.

➡ Result

The maximum number is 40

Note

- ◆ The `>=` operator is used so that equal numbers are also considered valid maximum values.
- ◆ The `&&` operator requires both comparisons to be true.

⊘ Warning

◆ The function must check all three numbers correctly. Using only one comparison, such as $\text{num1} > \text{num2}$, is not enough to find the maximum of three numbers.



Comparison

✓  num1 is maximum

- ✓ num1 $\geq$ num2.
- ✓ num1 $\geq$ num3.
- ✓ num1 is returned.

 num2 or num3 is maximum

- ✓ If num1 is not maximum, the program checks num2.
- ✓ If num2 is not maximum, num3 is returned.
- ✓ Only one result is returned.

✦✦ Summary

- ✓ Reference parameters allow readNumbers() to modify variables from main().
- ✓ The && operator combines two conditions that must both be true.
- ✓ if-else if-else can be used to find the maximum among three values.
- ✓ The >= operator handles equal values correctly.
- ✓ Functions separate input, calculation, and output into different tasks.

Thank you 😊